Summing and Difference Amplifiers: From Collected Data to Overleaf

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***Abstract*—This report investigates the operational performance of summing and difference amplifier circuits constructed using the LM741 operational amplifier and the NI myDAQ platform. Exercise 1 evaluates the summing amplifier across various resistor and voltage configurations, while Exercise 2 assesses the difference amplifier using dual AC signals. The experimental results for the summing amplifier demonstrated close agreement with theoretical calculations, with percentage errors ranging between 0.95% and 8.77%. Furthermore, the difference amplifier yielded a peak output voltage corresponding to an experimental voltage gain of 1.005, matching the ideal gain target. Overall, the findings successfully verified the theoretical operations and mathematical models of operational amplifier signal processing circuits.**

I. Introduction

A. Background and Objectives

Operational amplifiers are fundamental building blocks in analog signal processing, with summing amplifiers performing addition of input voltages and difference amplifiers subtracting two signals. Exercise 1 investigates how changing input and feedback resistor values alters the output voltage of a summing amplifier for fixed and variable DC inputs. Exercise 2 investigates a difference amplifier to evaluate its voltage gain by comparing the measured differential output to the theoretical input voltage difference $(V_2 - V_1)$. The integrated circuit used for both configurations is the standard eight-pin LM741 operational amplifier [3].

B. Equipment Used

- NI myDAQ
- myDigital Protoboard
- 741 Operational Amplifier IC (LM741)
- Resistors: 3.3 kΩ, 4.7 kΩ, 10 kΩ, 22 kΩ, 47 kΩ, 100 kΩ, 330 kΩ
- Multimeter leads and jumper wires

The complete experimental setup including the NI myDAQ and prototyping equipment is illustrated in Fig. 1.

II. Design

A. Exercise 1—Summing Amplifier

The summing amplifier circuit was constructed by powering the LM741 op-amp with ±15 V, grounding the non-inverting input (pin 3), and connecting feedback resistor R_f between pin 2 and pin 6. Input voltages V_1 and V_2 were applied through resistors R_1 and R_2 to pin 2, while the output voltage was measured at pin 6 using the myDAQ voltmeter. The complete setup is shown in Fig. 2, and the theoretical output voltage is defined by Equation 1:

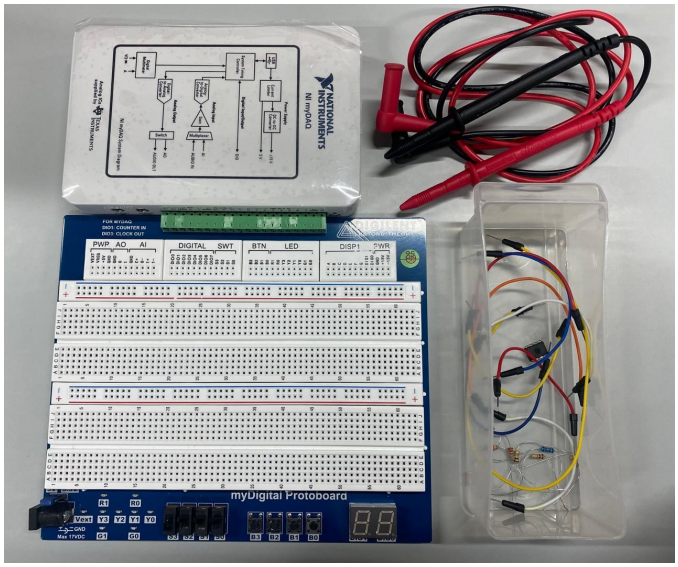


Fig. 1. Equipment used in the summing and difference amplifier experiments.

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right) \tag{1}$$

where V_{out} represents the output voltage, R_f is the feedback resistance, R_1 and R_2 are input resistances, and V_1 and V_2 denote the applied input voltages.

B. Exercise 2—Difference Amplifier

The difference amplifier was tested by applying two AC input signals ($V_1 = +5$ V$_p$ and $V_2 = -5$ V$_p$) to the circuit inputs. The output terminal at pin 6 was connected to the AI-0+ channel of the NI myDAQ to record the waveform. The peak-to-peak output voltage V_{pp} was then measured using the software oscilloscope to determine the experimental circuit gain.

III. Experimental Testing and Results

A. Exercise 1—Summing Amplifier

The supplied theoretical and measured results for the summing amplifier across three resistor combinations are com-

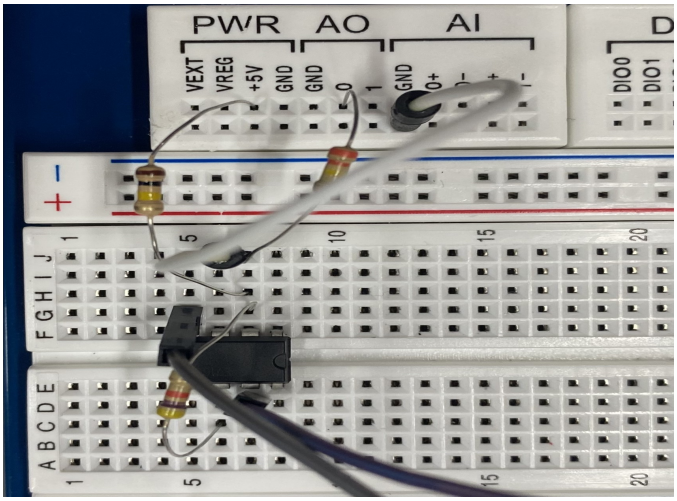


Fig. 2. Summing amplifier circuit connections.

piled in Table I., and the corresponding waveform is shown in Fig. 3. Across all evaluated trials, the measured output voltages closely tracked the calculated values, resulting in low percentage errors between 0.95% and 8.77%.

TABLE I
SUMMING AMPLIFIER RESULTS

R_f (kΩ)	R_1 (kΩ)	R_2 (kΩ)	V_1 (V)	V_2 (V)	Calculated V_{out} (V)	Measured V_{out} (V)	% Error
47	100	330	+5	+2	−2.635	−2.610	0.95
22	22	47	+5	−10	−0.319	−0.291	8.77
3.3	10	4.7	+5	+5	−5.161	−5.013	2.86

B. Exercise 2—Difference Amplifier

The measured performance data for the difference amplifier are presented in Table II. For differential AC inputs with $(V_2 - V_1) = 10.000$ V, the measured oscilloscope peak voltage V_{out} reached 10.049 V from a V_{pp} of 20.098 V. This yields an experimental voltage gain of 1.005, which is exceptionally close to the target unity gain of 1.

TABLE II
DIFFERENCE AMPLIFIER RESULTS

Calculated $V_2 - V_1$ (V)	Measured V_{pp} (V)	Measured V_{out} (V)	Gain $\frac{V_{out}}{V_2 - V_1}$
10.000	20.098	10.049	1.005

IV. CONCLUSION

This experiment successfully demonstrated the functional principles of LM741-based summing and difference amplifiers. The summing amplifier showed consistent agreement between theoretical predictions and measured DC outputs with errors under 8.77%. Additionally, the difference amplifier achieved an experimental gain of 1.005, confirming precise differential signal processing near unity gain. These findings fully satisfy the experimental objectives by validating theoretical circuit models using practical measurements.

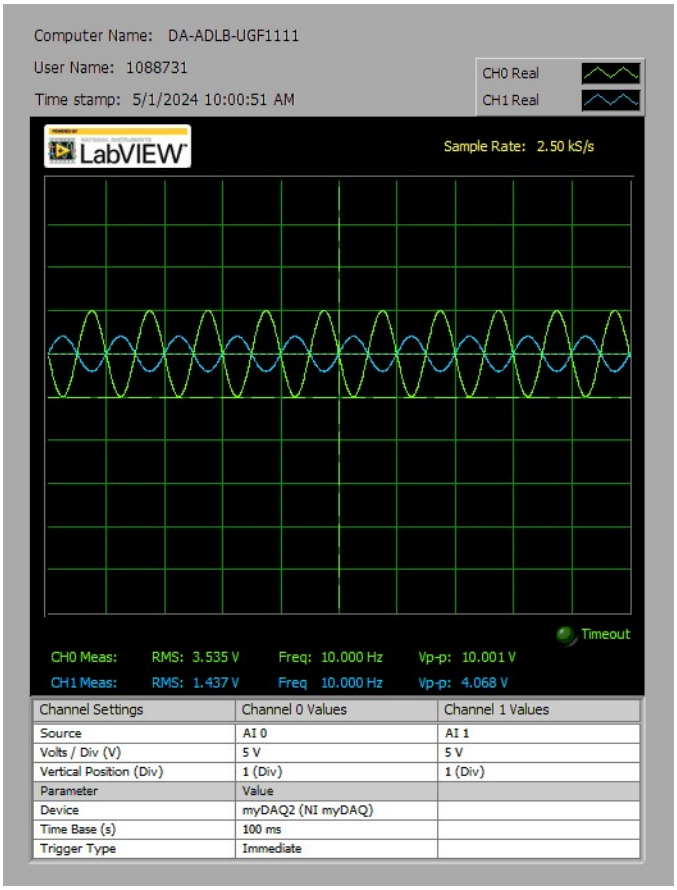


Fig. 3. Output signal displayed with the input signal on the oscilloscope.

REFERENCES

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